

Welcome to Introduction to Web Scraping!

Please log onto the Etherpad:

<https://pad.carpentries.org/2026-03-10-bham-intro-web-scraping>

Add your details in the 'Sign in' section

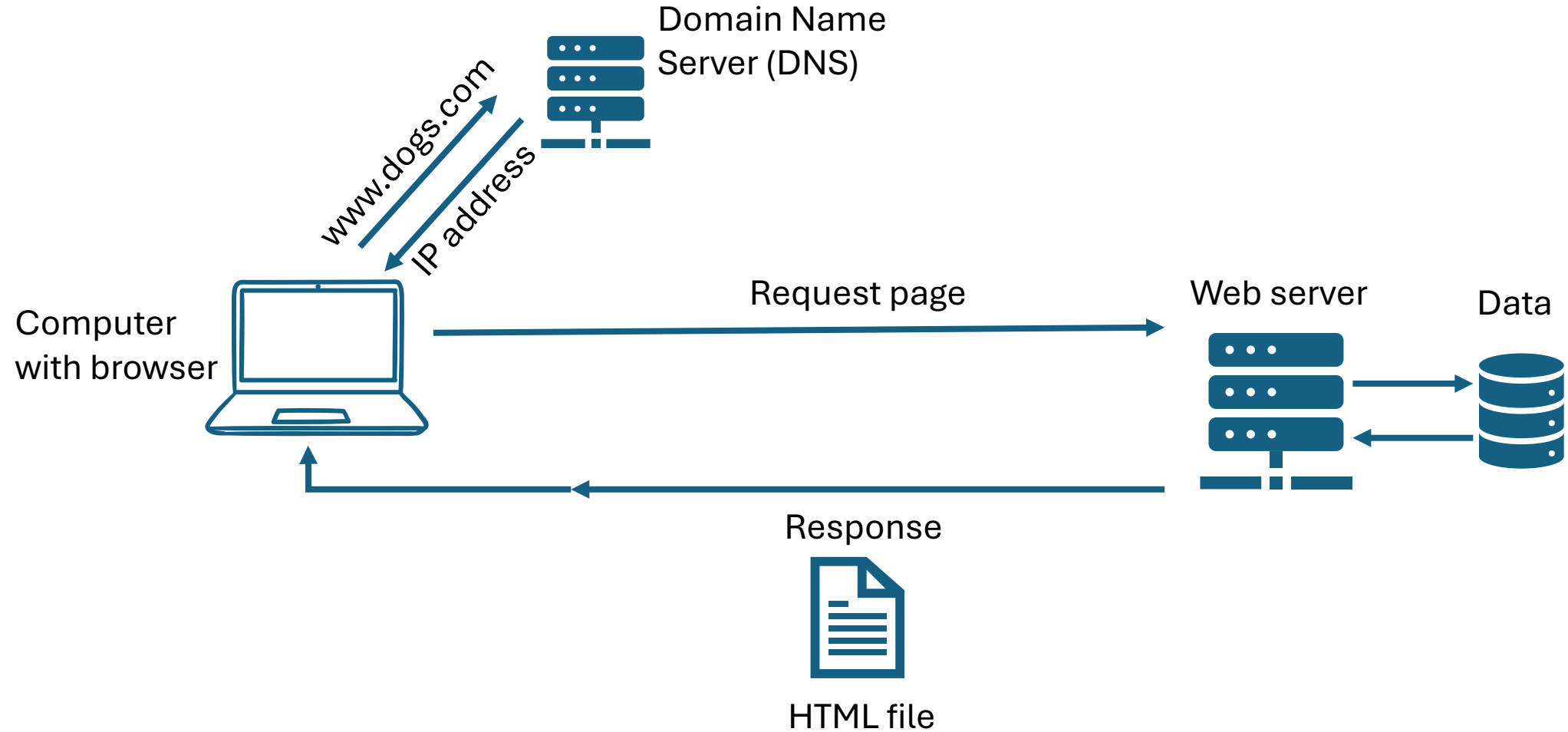
Check if you have completed the Software Setup'

“Web scraping (web harvesting or web data extraction) is a computer software technique of extracting information from websites.”

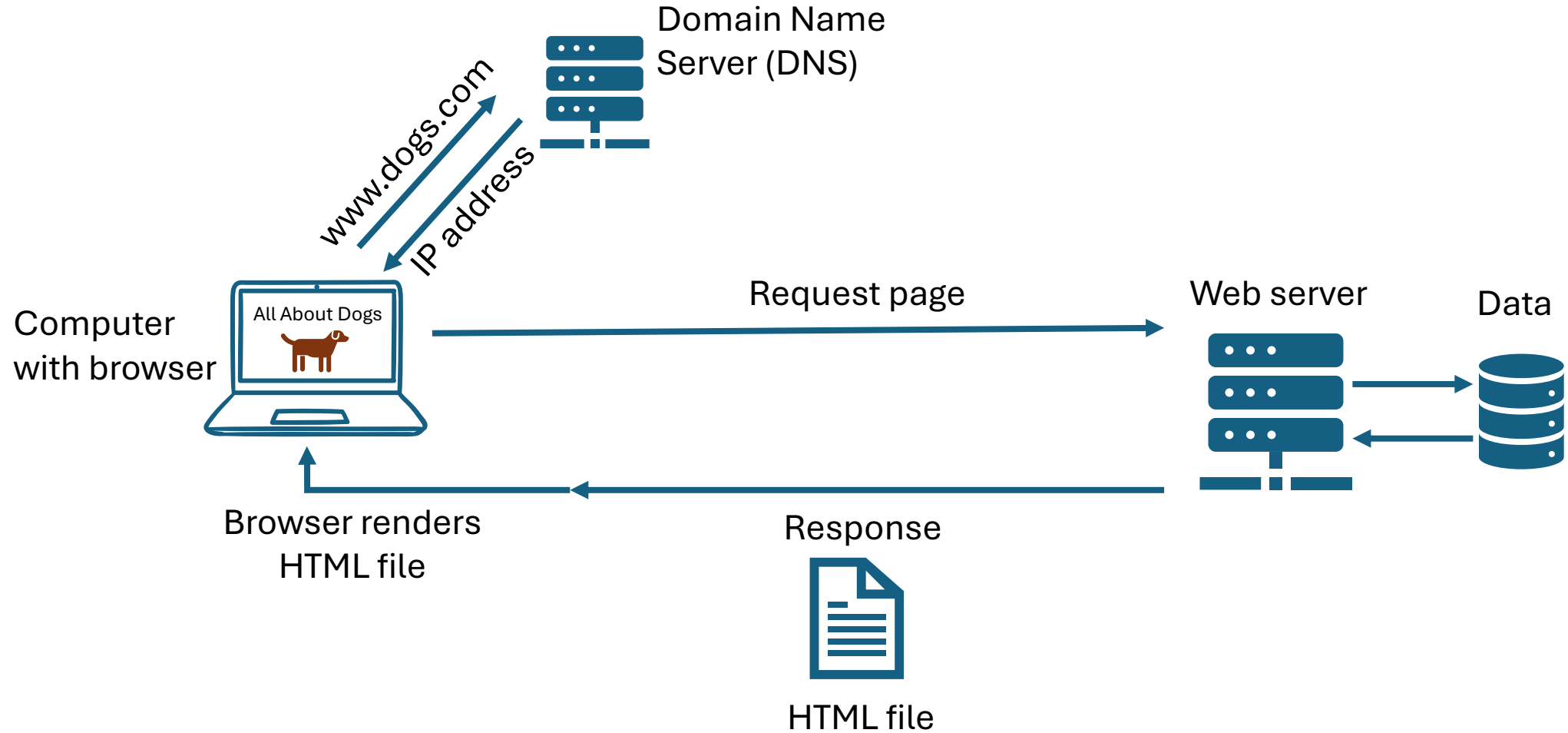
Wikipedia https://en.wikipedia.org/wiki/Web_scraping

Web scraping typically targets one web site at a time to extract unstructured information and put it in a structured form for reuse.

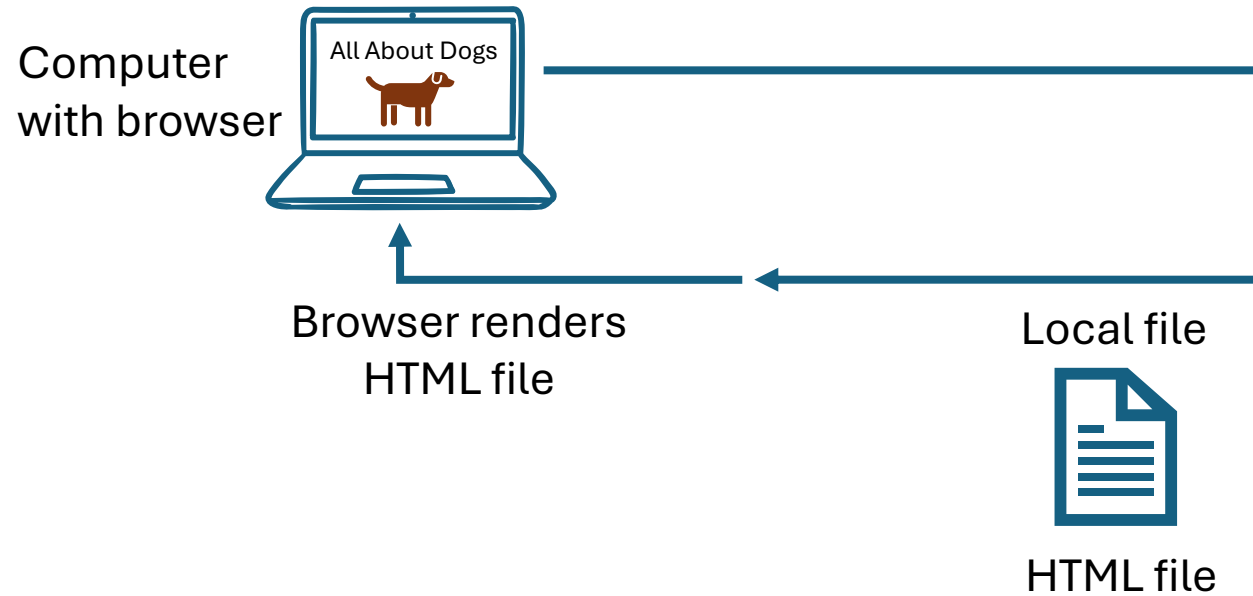
Requesting a Web Page



Browser Renders HTML

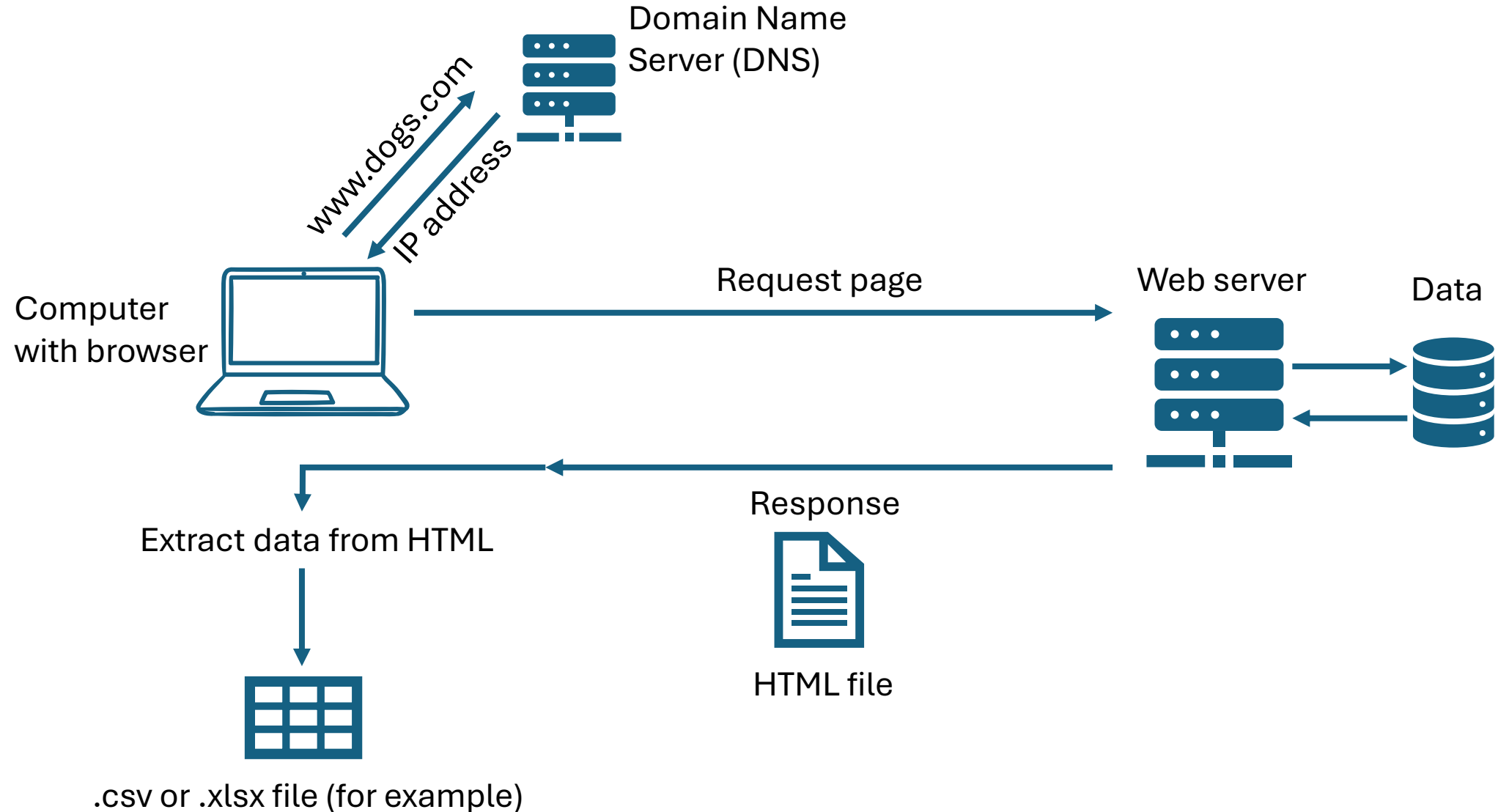


Browser Renders Local HTML File

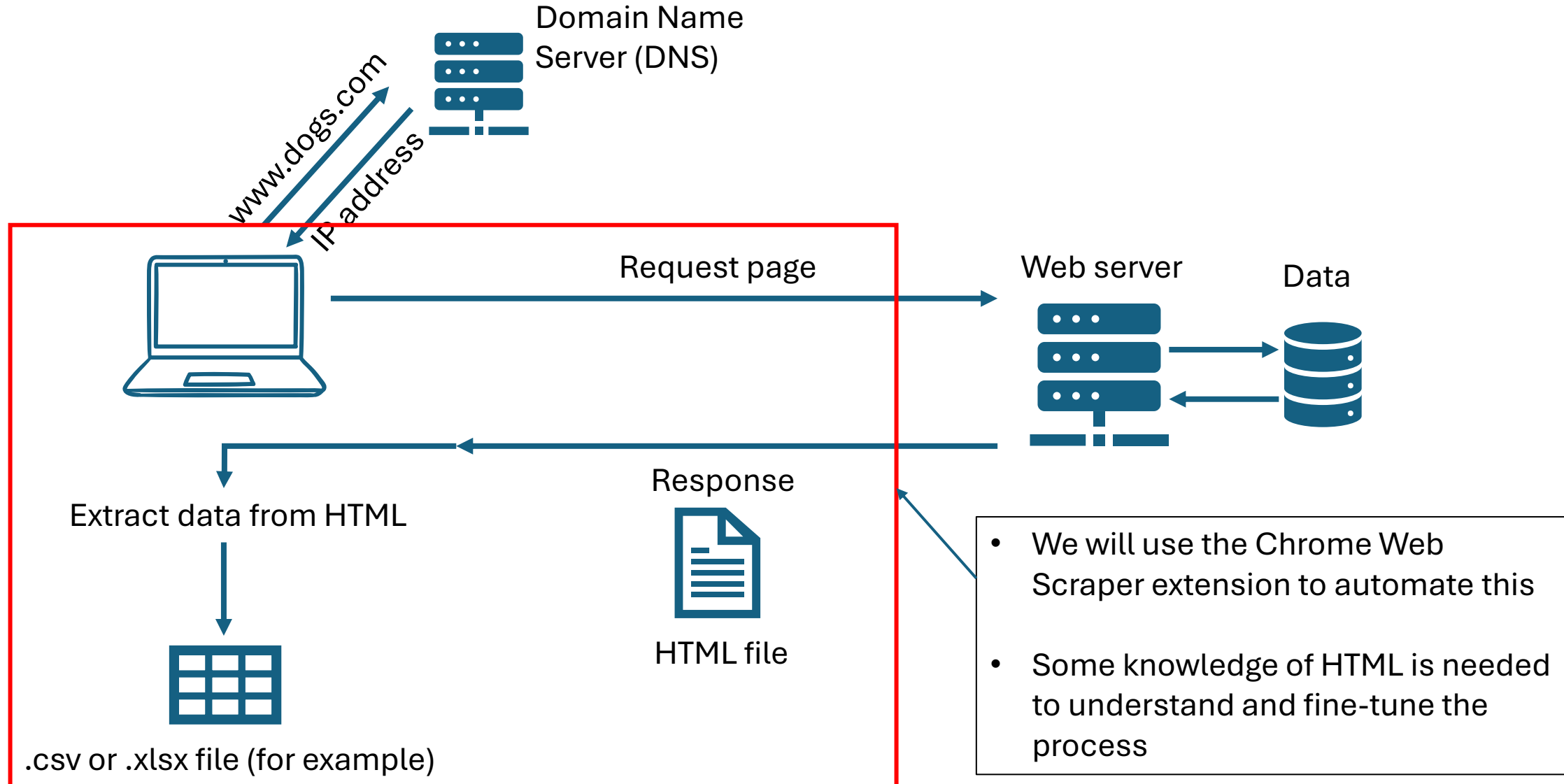


- An HTML file is just a text file with a specific format
- The browser can render a local file as well as fetching and rendering a remote one

Web Scrapping Process



Web Scraper Tool



html code from Canadian MPs webpage

(...)

```
<tr role="row" id="mp-list-id-25446">
  <td data-sort="Allison Dean" class="sorting_1">
    <a href="/members/en/dean-allison(25446)">
      Allison, Dean
    </a>
  </td>
  <td data-sort="Conservative">Conservative</td>
  <td data-sort="Niagara West">
    <a href="/members/en/constituencies/niagara-west(1124)">Niagara West</a>
  </td>
  <td data-sort="Ontario">Ontario</td>
</tr>
```

Web scraping might not be necessary

- It might be possible (and quicker) to copy and paste data into a spreadsheet (probably only for small amounts of data)
- Data may be available for download on a website
- An API (application programming interface) may be provided for extraction of data
- For much larger needs, [Freedom of information Act \(FOIA\)](#) requests can be useful. Be specific about the formats required for the data you want.

Hypertext Markup Language – HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>Sample web page</title>
</head>
<body>
  <h1>h1 Header #1</h1>
  <p>This is a paragraph tag</p>
  <h2>h2 Sub-header</h2>
  <p>A new paragraph, now in the <b>sub-header</b></p>
    <h1>h1 Header #2</h1>
    <p>
      This other paragraph has two hyperlinks,
      one to <a href="https://carpentries.org/">The Carpentries homepage</a>,
      and another to the
      <a href="https://carpentries.org/workshops/past-workshops/">past workshops</a>
    </p>
  </body>
</html>
```

Hypertext Markup Language – HTML

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```

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<head>
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This other paragraph has two hyperlinks,

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```

```
  </p>
```

```
</body>
```

```
</html>
```

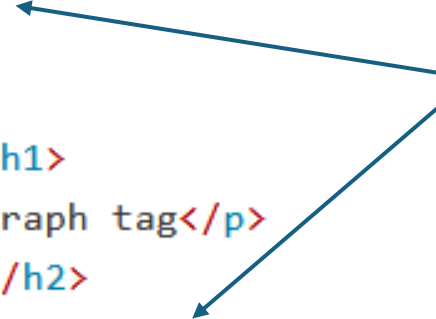
Elements marked by tags

All tags are in pairs
with a start <x>
and end </x> tag

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</html>
```

Text to be displayed
by the element
(between tags)



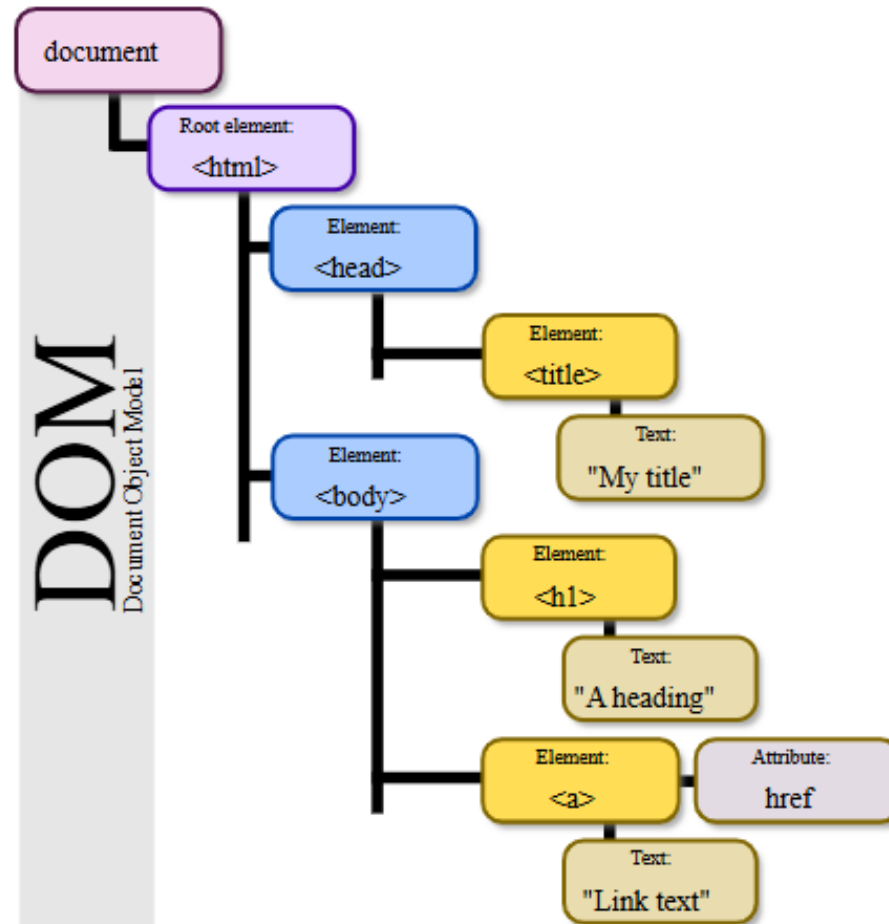
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</body>
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```

Attributes give more information.
Located inside the start tag:
name = "value"
Here a link with name 'href'

Text to be displayed
on the hyperlink

html Tree Format



Cascading Style Sheet (CSS) Code

```
<head>
  <title>Sample web page</title>
  <style>
    /* CSS Rule */
    h1 {
      color: blue;
      /* Property: value */
      font-size: 24px;
    }

    p {
      color: green;
      font-size: 16px;
    }
  </style>
</head>
```


Overall Structure of the Lesson Webpage

```
<!doctype html>  
<html lang="en" data-bs-theme="auto">  
  <head>  
    (...)  
    <title>Introduction to Web Scraping: Anatomy of a web page</title>  
    (...)  
  </head>  
  <body>  
    (...)  
  </body>  
</html>
```

Email address

```
(...)  
<div class="col-md-7">  
  <div class="contact-line">  
    <span class="label">Phone Number:</span>  
    <a href="tel:020 7219 4426">020 7219 4426</a>  
  </div>  
  <div class="contact-line">  
    <span class="label">Email Address:</span>  
    <a href="mailto:diane.abbott.office@parliament.uk"> diane.abbott.office@parliament.uk</a>  
  </div>  
</div>  
(...)
```

Selector: div.contact-line:nth-of-type(2) a